Swen360

Topic: Echo Eats

User stories and Tasks

1. As a consumer, I want to create an account with my email and password, so that I can reserve food. Registration should require a name, a valid email and a password of at least 8 characters. A duplicate email must be rejected with a clear message. Passwords should be hashed when shown, never in plain text. *As a customer, I would want to save my dietary preferences*.

   Tasks:

   - Wireframe the consumer sign-up screen
   - Create the users table and dietary preference fields
   - Build the registration endpoint with validation and password hashing
   - Build the sign-up form with inline error messages
   - Test valid, duplicate and invalid inputs

2. As a consumer, I would like to see a list of available items and filter by dietary tag, price and distance, so that I can quickly find something suitable. Only unexpired items with quantity above zero should be shown. *Each card should show a photo, a name, a vendor, the current price, and the original price, with the time left and distance*. The filters for dietary, maximum price, and maximum distance should work together. The results should be sorted by either distance, price or time left.

   Tasks:

   - Wireframe the "Deal List" screen and filter panel
   - Build the search endpoint with filters and sorting
   - Build the list view and filter controls
   - Test filter combinations and the empty-results state

3. As a consumer, I want to see an item's current price update on screen without refreshing, so that I always know what I would pay. *Current price = initial price * decay factor for time remaining*, and never falls below the vendor's minimum price. Open screens should refresh the price automatically (WebSocket or polling). *And The price shown at reservation time should be the price charged*.

   Tasks:

- Document the price formula and decay assumptions
- Implement the linear price calculation as a separate, testable module
- Implement the scheduled price-update job
- Build the live update channel to clients
- Show a live price and countdown on cards and the detail page

4. As a consumer, I want to reserve an item and have it held for 15 minutes, so that it is not sold to someone else while I complete payment. Reserving should reduce the available quantity immediately. A countdown should show the remaining hold-time. *When two users reserve the last unit at the same time: exactly one should succeed, the other sees "No longer available".*

   Tasks:

- Wireframe the reservation confirmation and countdown screens
- Create reservations table with state, price at reservation and expiry time
- Implement atomic stock reservation (transaction or row lock)
- Build reserve and cancel endpoints
- Build the reserve button, hold countdown and "My Reservations" screen